

Research: Automated Faceless Short-Form Video System (2026)

Status: settled research. Treat the facts here as decided unless verified otherwise. Verify anything marked **[VOLATILE]** before relying on it.

0. Executive summary

- Viable path: a portfolio of 4-12 **faceless, template-able** accounts built on royalty-free footage (Pexels/Pixabay CC0) with **original narration and captions**. Not reposting.
- Two hard blockers in the original plan: 1. TikTok's Content Posting API only posts **private** videos until a manual audit passes (2-6 weeks). 2. Free-tier hosting struggles with scheduled CPU video rendering. GitHub Actions is workable; Oracle Cloud Always Free ARM is the real answer.
- Monetisation is realistically 3-18 months out and gated on "originality" rules that specifically target mass-produced minimal-edit content. Build for fun now; add transformation to monetise later.

1. Viral content landscape

1.1 Macro numbers

Metric	Value	Source / date
YouTube Shorts daily views	200B+	Neal Mohan, Cannes Lions, Jun 2025 (up 186% from 70B in Mar 2024)
TikTok time/day (global avg)	~95 min	FT, Q4 2023
TikTok time/day (US adults)	~52 min	eMarketer, Apr 2025
Reels share of IG time	~50%	Meta
Engagement rate	TikTok ~2.8% · Reels ~0.65% · Shorts ~0.30%	2024 baselines
Multi-platform reach multiplier	4-5x vs single platform	industry reports

"Brain rot" was Oxford's 2024 Word of the Year (Dec 2, 2024; 37,000+ votes; +230% usage YoY).

[VOLATILE] Some commentary argues absurdist brainrot peaked in 2025 and 2026 is seeing a partial "intellectual revival" (Substack, longer content, BookTok). Treat absurdist formats as high-variance and rotate them.

1.2 Active niches

- Italian Brainrot** (Tralalero Tralala, Bombardiro Cocodrilo, Tung Tung Tung Sahur) — surreal animal/object hybrids, pseudo-Italian rhyming narration. Canonical versions are AI-generated, but the *format* (absurd recurring character + nonsense rhyme + caption) is imitable with stock or illustration.
- "6 7" / "67"** — nonsense catchphrase, big with Gen Alpha.
- Skibidi Toilet** (DaFuq!?Boom!) — meme-fuelled animated series.
- Low-prop participation trends: "parents reading brainrot words", "propaganda I'm not falling for", "the chair".

1.3 Faceless account benchmarks

Account	Scale	Format
Noah Morris (18 channels)	~2.5M subs total	faceless automation, sports/crime/celebrity
The Infographics Show	~15M subs	animated explainers
WatchMojo	~26M subs	list videos
"Am I the Jerk?"	1.2M+ subs	Reddit AITA narration + simple visuals
@texty.stories.daily	500k+ followers, 25M+ likes	text-based videos

Reddit-story channels commonly reach 100k-1M+ subs within a year; individual videos cross 5M views with 65-80% watch-through.

1.4 Automatability tiers

Highly automatable — Reddit story → TTS → gameplay background + captions · satisfying loops · ranking/"top 5" · text-POV / greentext cards · fact bombs over B-roll · comparison videos.

Semi-automatable — meme caption over clip (needs taste + safe source) · reaction-style (needs commentary to be transformative) · trend-jacking (needs speed + judgement).

Not automatable — original skits, opinion/commentary with a voice, character-driven series.

1.5 Candidate concepts

#	Concept	Automatable	Raw material	Originality risk
1	Reddit stories over gameplay	very high	Reddit text + free-to-use gameplay	HIGH — the canonical "minimal transformation" target
2	Oddly-satisfying loops	high	Pexels/Pixabay CC0	MEDIUM — saturated
3	Ranking / Top 5 over B-roll	high	stock + LLM script	MEDIUM
4	Greentext / text-POV cards	very high	LLM or curated text	LOW-MEDIUM
5	"Did you know" fact bombs	high	stock B-roll + LLM	LOW-MEDIUM
6	Original absurdist character series	medium	illustration / CC stock	LOW
7	Quotes over cinematic stock	very high	stock + music	MEDIUM — saturated

7	Quotes over cinematic stock	very high	stock + music	medium	screen-recorded
8	Micro-tutorials / life hacks	med-high	stock or screen record		LOW
9	"This vs That" comparisons	high	stock + data		LOW-MEDIUM
10	Screen-recorded internet finds	medium	needs commentary		MEDIUM

Launch set: #1, #2, #4, #5 — highest automatability, no filming, defensible originality provided narration/captions/edit are your own.

2. Platform policy reality

2.1 YouTube — "inauthentic content", July 15, 2025

YouTube updated guidelines "to better identify mass-produced and repetitious content". Creator Liaison Rene Ritchie framed it as "a minor update to YouTube's longstanding YPP policies". Explicit target: channels uploading narrated stories with only superficial differences, or slideshows sharing the same narration. This content was already ineligible for monetisation.

No change to the reused-content policy — commentary, clips, compilations and reaction videos remain permitted. Transformation is the dividing line. AI is not banned; AI-assisted content is monetisable if it carries unique human value.

A July 2026 clarification added "unsatisfying/offputting content" and "AI personas related to sensitive topics".

2.2 TikTok — Originality Policy

Unoriginal content (copied, or with "minimal original input or edits") is ineligible for the For You feed and for Creator Rewards, where originality is a scoring input. Reported enforcement triggers: deleting and reposting your own video, compilations, screen recordings, slideshows, lightly-edited re-uploads. "Low quality" (split screens, meaningless reactions, slideshows) is explicitly called out. Repeated flags can permanently revoke monetisation.

2.3 Instagram

Penalises aggregators/reposters; down-ranks watermarked content from other apps. API publishing cap cited as both **25** and **100** posts/24h — plan against 25. Post 1-3 Reels/day for algorithmic trust.

2.4 AI disclosure

Platform	Requirement	Applies to this project?
TikTok	C2PA Content Credentials since Jan 2025. Label required for synthetic faces/voices/realistic scenes. AI-assisted text (scripts, captions, hashtags) is exempt.	Mostly no
YouTube	Disclose "realistic altered or synthetic content". Standard editing/text/AI-assist doesn't trigger.	No
Meta/IG	"AI info" labels via C2PA.	No

TikTok's Fifth Transparency Report: 51,618+ synthetic media videos removed in H2 2025 (+340% YoY); 1.3B+ AI-generated videos labelled.

Since we generate no AI video, disclosure mostly doesn't apply. TTS narration over non-realistic footage generally does not require the synthetic-media label — but does if you clone a real voice.

2.5 What actually happens

Bans are rare for benign meme content. Real consequences are: (a) demonetisation / rewards ineligibility, (b) reduced distribution ("shadowban"), (c) removal only for copyright or community-guideline violations.

Aggregators survive by not depending on platform ad revenue (shoutouts, affiliates, driving to a product) and by adding transformation.

2.6 Copyright

- **YouTube Content ID** scans every upload; matches block, mute, or divert monetisation.
- **TikTok** has its own audio/copyright detection; a track cleared on TikTok may be blocked on Reels.
- **"Transformative"** in practice = substantial new expression (narration, commentary, an edit that changes purpose). Captions on someone else's clip is not transformative.
- Fair use is a legal defence, not protection from takedown. Platforms enforce more conservatively than courts. 3 YouTube strikes = termination.

2.7 Automated posting

Permitted via the sanctioned APIs after their reviews. Unofficial automation and watermark-removal repost tools violate ToS.

3. Publishing APIs

3.1 TikTok Content Posting API

- Needs a developer app **plus** a separate Content Posting audit.
- **Unaudited apps post SELF_ONLY (private) only.** Account must be private at post time; owner manually flips account and each post to public. True unattended public posting is impossible pre-audit.
- Unaudited caps: 5 users / 24h; **6 requests/min** per user token. Access token 24h, refresh token 365 days.
- Audit: typically 2-6 weeks (some report 1-2). Clear documented use cases (schedulers, dashboards) get approved; vague ones get rejected, adding 1-2 weeks each.
- **No native scheduling parameter** — publish now or draft. Build your own queue.
- Video only, no photo carousels.

- Two modes: **Direct Post** (needs audit for public) and **Creator Draft** (lands in TikTok inbox for a human to publish — works without audit).

3.2 YouTube Data API v3

- **[VOLATILE]** Dec 4, 2025 revision: upload quota cost dropped from ~1600 units to **~100 units**; default allocation **~100 videos.insert calls/day** in a dedicated bucket separate from the shared 10,000-unit pool (as of ~Jun 2026). Many 2026 guides still quote 1600 — confirm in your own Cloud Console.
- Default 10,000 units/day per project, resets midnight PT. `search` costs 100 units each — avoid.
- **Unverified apps upload as private** until the Audit and Quota Extension passes (manual review, weeks-months).
- No monetary cost; quota cannot be purchased, only audited for.

3.3 Instagram / Meta Graph API (Reels)

- Requires **Instagram Business** account (Creator accounts are NOT supported — common gotcha), linked Facebook Page, Meta developer app, and approved `instagram_business_content_publish` (replaced older scopes Jan 27, 2025).
- App review 2–4 weeks per submission, with a screencast of the full flow.
- Container model: `POST /{ig-user-id}/media (media_type=REELS, public video_url )` → poll container status until FINISHED → `POST /{ig-user-id}/media_publish`.
- Media must sit at a **publicly accessible URL** at post time.
- Reels eligibility: 9:16, 5–90s, H.264/HEVC.
- Rate limit 25 (some sources 100) posts/24h; Reels+Stories+Feed share the bucket.

3.4 Third-party posting layers

Postiz — open source (AGPL-3.0), self-hostable with no feature gap vs hosted, ~29–32k GitHub stars. Supports 30+ platforms including TikTok, Instagram, YouTube, Facebook, Reddit, Threads, X, Pinterest, Bluesky, Mastodon, Discord. Has a public API, NodeJS SDK, n8n node, Make.com integration, and an **MCP server**. Self-hosted is genuinely free.

It uses official platform OAuth, so it does **not** bypass TikTok's audit or Meta's app review — but it removes the burden of implementing each API, chunked uploads, token refresh, and multi-step flows. Paid tier channel counts: Standard 5, Team 10, Pro 30, Ultimate 100.

Alternatives: Upload-Post (unlimited posts, tight profile limits), Ayrshare (per-profile), Blotato, Post for Me (~\$10/1,000 posts), Zernio/PostEverywhere. Pre-audited platforms are effectively "rent someone else's approved TikTok app" — fastest route to public posting.

Decision: use self-hosted Postiz as the publishing layer.

4. Sourcing

4.1 Recommended sources

Source	License	Limits
Pexels API	Pexels License (CC0-style), commercial OK, no attribution	200 req/hr, 20,000/mo (liftable free with attribution)
Pixabay API	Pixabay License, commercial OK, no attribution	unlimited advertised; must download to your own server, no permanent hotlinking
Videvo / Mixkit / Coverr	per-clip tiers — check	—
Internet Archive	public domain (verify per item)	good for retro/absurdist montage
Wikimedia Commons	PD/CC, needs attribution handling	rich metadata

Primary source: Pexels. Secondary: Pixabay.

4.2 Reddit

Posting to Reddit does **not** transfer copyright — the uploader (or a third party) still owns it. Reddit's user agreement grants Reddit a license, not you. Using Reddit *video* without permission is exactly what gets flagged and DMCA'd. Reddit *text* is lower risk (still credit; some subs require permission) but must still be transformed via your own narration and captions.

4.3 Gameplay backgrounds

Minecraft parkour / Subway Surfers / GTA loops are widely published as free-to-use for exactly this purpose. Use clips explicitly labelled free-to-use / no-copyright, or record your own.

4.4 Trend discovery

No clean official API. Practical: TikTok Creative Center, YouTube trending pages, Reddit API (`/r/<sub>/top`), tools like Virlo. Scraping TikTok/YouTube beyond official APIs violates ToS.

4.5 yt-dlp — stated plainly

yt-dlp is legal open-source software (Unlicense). Downloading from YouTube **violates YouTube's ToS** — a contract matter (remedy: ban or civil action), not automatically criminal. The real exposure is **copyright**: re-publishing others' video without permission is infringement (US statutory damages up to \$150k/work if wilful). Safe uses: your own content, public domain, explicit CC with attribution.

Also technically fragile: YouTube's 2025 anti-download measures (PoToken/SABR) frequently break yt-dlp, requiring cookies/token-provider plugins.

Do not build the pipeline on yt-dlp scraping. Use stock APIs.

5. Rendering

5.1 Library choice

- **MoviePy 2.0** — current, and it broke the v1 API completely. No compatibility shim. Effects moved from functions to classes. Import path is now `from moviepy import ...` (not `moviepy.editor`). `TextClip` requires an explicit `font=` path to a real OpenType file on disk — **on Linux there is no system font resolution, you must ship a .ttf**. Renames: `.set_position` → `.with_position`, `.set_duration` → `.with_duration`.
- **ffmpeg-python** — thin Pythonic wrapper; fine for transcode/filters.
- **Direct FFMpeg subprocess** — most reliable and fastest. Most production pipelines end up here. **This is our primary path.**

5.2 Captions

faster-whisper (CTranslate2 reimplementation of Whisper, ~4x faster, lower memory, runs on CPU with `--device cpu`).

Pipeline: FFMpeg extracts audio → faster-whisper transcribes with word timestamps → generate `.ass` → FFMpeg burns in with `-vf "ass=captions.ass"`.

For word-by-word "brainrot captions": ASS file with `PlayResX=1080`, `PlayResY=1920`, white text, thick black outline (outline 7-9), `margin-v` 120-180 to clear platform UI.

On CPU use `base` or `small`. Whisper `turbo` (~8x faster than large-v3) is the sweet spot with GPU.

5.3 Resource envelope

A 30-60s 1080x1920 render with burned captions on CPU: **~1-5 minutes** depending on cores and complexity. faster-whisper on `base / small` transcribes ~1 min of audio in well under a minute on CPU. RAM ~1-2 GB baseline, spikes with MoviePy compositing + Whisper.

Target: ≥2 GB RAM, 2 vCPU. No GPU required. Fits GitHub Actions runners (7 GB / 2 vCPU) and Oracle ARM comfortably.

5.4 Audio

Source	License	Note
Pixabay	Pixabay License, commercial OK, no attribution	primary — music + SFX
Mixkit	free SFX/music	
Freesound	per-sound CC — some CC0, some require attribution	check each
YouTube Audio Library	free, some require attribution	
Uppbeat / Free Music Archive	per-track terms	

"Royalty-free" ≠ cleared on every platform. Prefer CC0/Pixabay and bake pre-cleared audio into the render — platform-native audio libraries are not available via API.

6. Hosting

Workload: CPU-heavy render + network I/O, ~3 runs/day per account.

Option	Verdict
Oracle Cloud Always Free (ARM Ampere A1)	Best. Up to 4 OCPU / 24 GB RAM free forever. Real always-on VM, real cron, install ffmpeg. Caveat: notorious "out of capacity" for ARM in popular regions — use a creation-retry script.
GitHub Actions	Best zero-setup free option. 2,000 min/mo private, unlimited on public repos. <code>schedule:</code> cron, 6h job limit. ffmpeg installable. Caveats: cron is imprecise (15-45+ min delays) and scheduled workflows auto-pause after 60 days without a push. ~35 min/day for 3 accounts fits easily.
Google Cloud Run Jobs	Good serverless route. Generous free tier, Cloud Scheduler for cron, ffmpeg works in your container. Watch egress.
Fly.io / Heroku	No real free tier anymore.
Railway / Render	Credit/trial based; Render free has cold starts and limited hours. Built-in cron but constrained CPU.
PythonAnywhere	Free tier is daily only (not every 8h) + manual 3-month renewal. Unsuitable.
Hugging Face Spaces	Demo-oriented, sleeps, ToS mismatch. Poor fit.
Replit	Always-on requires paid.

Recommended architecture: one Oracle ARM VM running (a) the Python render script on cron and (b) self-hosted Postiz. All free. GitHub Actions as fallback if Oracle capacity is unavailable.

7. LLM choice

7.1 Xiaomi MiMo

Family launched Apr 2025 (MiMo-7B). V2 series deprecated ~Jun 30, 2026. Current API lineup:

Model	Params	Context	Price (in/out per M)
MiMo-V2.5-Pro	~1.02T total / 42B active (sparse MoE)	1M	\$0.435 / \$0.87
MiMo-V2.5	310B total / 15B active, omnimodal (text/image/video/audio)	1M	\$0.14 / \$0.28
MiMo-V2.5-ASR	speech recognition	—	—
MiMo TTS series	[VOLATILE] free for a limited time	—	—

Prices reflect a May 27, 2026 cut. Historical figures that still circulate: \$1/\$3 was the deprecated **V2-Pro**; \$0.10/\$0.30 was **V2-Flash** at its Dec 16,

2025 launch.

Open weights (MIT): MiMo-7B, MiMo-VL-7B (vision), MiMo-Embodied-7B, V2.5 weights. Self-hostable via vLLM/SGLang (Xiaomi maintains a vLLM fork; V2.5 has an official SGLang FP8 cookbook). Downloads at huggingface.co/XiaomiMiMo.

Access: official console platform.xiaomimimo.com (login via Xiaomi ID; `sk-...` key). Base URL <https://api.xiaomimimo.com/v1> (OpenAI-compatible) and [/anthropic](#) (Anthropic-compatible). Supported by LiteLLM as `xiaomi_mimo/<model>`. **[xiaomi-mimo-ai.com](#) and [mimo-v2.com](#) are unofficial — do not get keys there.** Also on OpenRouter (~\$0.14/\$0.28 for V2.5, ~\$0.35/\$0.70 for V2.5-Pro).

MiMo-7B runs on ~8 GB+ RAM (INT4 quant available) — could run on the Oracle ARM VM.

7.2 Comparison for this task

Option	Cost	Vision	Verdict for this project
Gemini free tier (2.x Flash)	free	yes (image + video)	Best default. Free, multimodal, well documented.
Xiaomi MiMo V2.5	\$0.14/\$0.28	yes (omnimodal)	Fine and very cheap. Use if you want MiMo.
Groq	free/cheap	limited	Extremely fast for high-throughput caption/hook generation.
DeepSeek	very cheap	no	Strong text/reasoning.
Qwen	cheap / self-host	yes (Qwen-VL)	Solid alternative.
Local via Ollama	free	varies	Zero marginal cost on the Oracle VM, no rate limits, private. Slower/weaker but fine for captions and hooks.

Captions, hooks, hashtags and concept selection are easy text tasks — any of these is capable. At 3 videos/day across a few accounts, token spend is negligible regardless of choice.

Verdict: Gemini free tier as primary, local Ollama as zero-cost fallback, MiMo as a supported option. There is no capability reason MiMo is required — it's a cost-justified preference. Keep the LLM behind an interface so this is a config change.

8. Rollout plan

Stage 0 — model the problem. "Single script, fully unattended, all 3 platforms, free hosting" is achievable but not on day one, because of the TikTok audit and Meta app review. Also: prefer a small modular pipeline over one script — far easier to debug and to run per-account.

Stage 1 — build and test locally (week 1-2). Pick 4 concepts (#1 Reddit-story, #2 satisfying loops, #4 text-POV, #5 fact bombs). Build source → script → TTS → render → captions. Render locally, measure time and quality.

Stage 2 — publishing (week 2-4). YouTube first (easiest API; complete the audit). Instagram next (Business account + Page + app review). TikTok via Creator-Draft or Postiz while your own audit is pending.

Stage 3 — deploy (week 3-4). Oracle ARM VM: ffmpeg + Python + Postiz + cron every 8h. GitHub Actions if Oracle capacity fails.

Stage 4 — scale and measure. One account per concept, 3 posts/day for 3–4 weeks. **Kill/keep after ~30 posts:** keep if median >2,000–5,000 views or any post >50k. Kill if median <500. Watch for unoriginal-content flags as a transformation signal.

Stage 5 — monetisation (month 3+). Only for accounts that cleared the view threshold *and* pass originality. Diversify beyond ad revenue (affiliate, shoutouts, own product) — ad revenue is the most originality-gated.

Thresholds that change the plan: - TikTok audit rejected twice → commit to Postiz/third-party or Creator-Draft only. - Render >10 min/video or RAM ceiling hit → smaller Whisper model, pre-cache stock, move to Oracle. - "Inauthentic/unoriginal" strike → increase transformation immediately or retire the concept.

9. Caveats

- **[VOLATILE] numbers.** The YouTube quota change (1600 → ~100 units, Dec 4 2025; dedicated upload bucket ~Jun 2026) is in Google's official revision history, but 2026 guides still cite 1600 — confirm in your Cloud Console. Instagram's per-day cap is cited as both 25 and 100 — verify via the `content_publishing_limit` endpoint.
- **MiMo churn.** Lineup and prices changed several times in 12 months. Verify at platform.xiaomimimo.com before committing. Some performance claims are Xiaomi marketing.
- **Originality enforcement is opaque.** False-positive flags on 100% original content are reported; appeals are inconsistent. Treat monetisation as uncertain.
- **Third-party posting tools track shifting APIs.** Postiz maintainers describe TikTok/Instagram integrations as "a moving target".
- **Legal.** Reposting others' clips and yt-dlp scraping both violate ToS and risk copyright liability. Stock-API + original narration is the defensible path.
- **Not legal advice.** If revenue becomes material, consult a media/IP attorney.